Design of a Server Authoritative Multiplayer Architecture

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Abstract –

Real time multiplayer systems face critical challenges in ensuring fairness. High Latency, Packet Loss and malicious client behaviour end up degrading user experience. Hence, an architectural solution is required to fix the above by trying to maximise performance and security without comprising either. This task is very hard as performance (speed) and safety are generally inversely dependant attributes.

This project proposes usage of a Client Server Architecture with Server Autorotative structure. The singular main server will act as a source of truth, while the clients are just acting as input providers. This design ensures that one client itself cannot degrade the gameplay experience of other users. Thus, the core idea is to ensure safety. To tackle reduced performance, various techniques like lag compensation, client-side prediction etc will be used.

The prototype will focus on a lightweight 2D grid-based environment where multiple clients can connect, move entities, and perform actions in real time. The scope includes networking, synchronization, and persistence of player data, but excludes advanced graphics, large scale deployment, and high availability clustering due to resource limitations.

This project is relevant for architectural comparison because it illustrates how quality attributes such as performance, fairness, modifiability, and security can be achieved through specific design decisions. It provides a clear case study of trade-offs, such as prioritizing low latency and consistency over availability tactics like replication or fault tolerance, which are impractical in a constrained academic setting.

Marketing Architecture Diagram –